Speaker Information

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Talk Information

The neural circuit basis of flexible action policies.

Intelligence requires computing a policy for action by integrating information represented internally with that encoded in sensory populations. To determine the circuit mechanism of this computation, we examine cortical processing during smooth pursuit eye movements in two areas critical to the behavior: area MT, where neurons encode visual motion information and the smooth pursuit eye movement region of the frontal eye fields (FEFsem), where neurons flexibly modulate pursuit behavior. We determine how MT inputs are transformed into an action policy embedded in the FEFsem population response by recurrent circuit connections. Beyond simply integrating MT inputs, the circuit implements interactions between internal representations critical to the computation of the action policy. Together, the results connect sensory drive, individual neuron responses, and circuit dynamics to the formation and manipulation of internal representations critical to achieving intelligent behavior.

Research Overview (for collaboration discussion)

[Research interests] Cognition, sensory-motor behavior, neural circuits, dynamical systems

[Current Research Projects] Neural circuit basis of flexible action policies, Neural implementation of mental simulation

[Techniques] Psychophysics, neurophysiology, computational modeling, recurrent neural networks

[Aspects or techniques of interest for future collaboration] High dimensional neural recording and stimulation, neural circuit segmentation, multi-area interconnectivity and processing, learning and development, brain-computer interface

Short Bio

I study the neural mechanisms underlying complex behavior using experimental and computational approaches in humans and nonhuman primates. I received my PhD in Neuroscience with Kenneth Britten at UC Davis and then went on to postdoctoral positions with Mehrdad Jazayeri and Stephen Lisberger at MIT and Duke University, respectively.